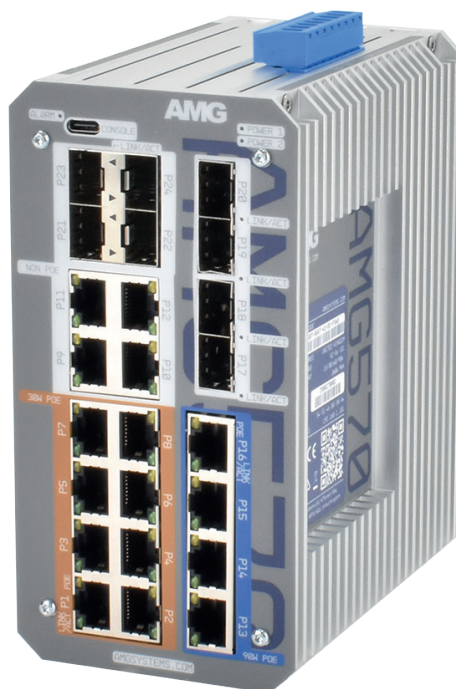


# AMG570 SERIES 16-24 PORT INDUSTRIAL GRADE MANAGED LAYER 2+ SWITCHES WITH OPTIONAL 30/90W POE



## Industrial Ethernet Solutions

AMG's fully managed layer 2+ Ethernet switches provide 100Mbps, Gigabit and 2.5 Gigabit Ethernet switching for industrial network applications. Available with 8, 12 or 16x RJ45 Gigabit ports supporting optional 30/60/90W PoE, 2, 6 or 10x 100Mb/1Gb SFP ports and 2x 100Mb/1Gb/2.5Gb SFP ports.



|                      |                     |                      |                |                       |                          |                     |                         |                    |                |                     |                       |
|----------------------|---------------------|----------------------|----------------|-----------------------|--------------------------|---------------------|-------------------------|--------------------|----------------|---------------------|-----------------------|
|                      |                     |                      |                |                       |                          |                     |                         |                    |                |                     |                       |
| Gigabit<br>up to x16 | Gigabit<br>up to x4 | Gigabit<br>up to x16 | 1/2.5 Gb<br>x2 | 100/1000<br>up to x10 | Contacts<br>1 In / 2 Out | Temp<br>-40°C~+75°C | Mounting<br>DIN/Surface | Protection<br>IP40 | PSU<br>x2 Dual | DANTE®<br>Supported | TAA/BABA<br>Compliant |

### / OVERVIEW

AMG570 series layer 2+ managed industrial Ethernet switches are designed in a thermally efficient DIN rail or wall mount housing and have 8, 12 or 16 Gigabit Ethernet RJ45 ports with an additional 4, 8 or 12 multi-rate SFP ports that support 100Mb and 1Gb speeds. Two of these SFP ports also support 2.5Gb for high speed data uplink into core networks, providing application and site flexibility.

Available as non-PoE models as well as multiple PoE models for IEEE802.3at 30W and/or IEEE802.3bt 60/90W they are suitable for powering the latest high powered PoE devices over a wide industrial operating temperature range.

The AMG570 series support a wide range of management functions as well as Rapid Spanning Tree Protocol (RSTP), Multiple Spanning Tree Protocol (MSTP), Media Redundancy Protocol (MRP) and Ethernet Ring Protection Switching (ERPS) for network redundancy. IGMP functionality is supported to handle the multicast traffic which is commonly used in IP CCTV deployments. Dante® audio networks are also fully supported.

SFPs and PSUs need to be ordered separately.

### / FEATURES

- Innovative thermally efficient housing ensures high levels of device reliability under full load
- -40°C to +75°C temperature maintains performance in extreme conditions
- DIN rail or wall mountable - quick to install and remove for maintenance
- Compliant with all IEEE 802.3 speeds (i/u/ab/z)
- Dual or Triple Speed SFP ports (100Mb, 1Gb, 2.5Gb)
- Supports RSTP, MSTP, ERPS, MRP, SNMP and IGMP
- IEEE 802.1x port security enabled
- Designed for industrial Dante® audio networks
- Supports optional 15W, 30W, 60W and 90W PoE
- Layer 3 Static Routing†
- Designed in the USA & UK. Manufactured in the United Kingdom or USA (BABA Compliant)
- AMG Lifetime Support Warranty

# Specifications.

## Standards.

|                                |                                           |
|--------------------------------|-------------------------------------------|
| IEEE 802.3i                    | 10Base-T                                  |
| IEEE 802.3u                    | 100Base-TX & 100Base-FX                   |
| IEEE 802.3ab                   | 1000Base-T                                |
| IEEE 802.3z                    | 1000Base-X                                |
| IEEE 802.3x                    | Flow Control                              |
| IEEE 802.3ad                   | Port Trunk with LACP                      |
| IEEE 802.3az                   | Energy Efficient Ethernet                 |
| IEEE 802.1D                    | Spanning Tree (STP)                       |
| IEEE 802.1w                    | Rapid Spanning Tree (RSTP)                |
| IEEE 802.1s                    | Multiple Spanning Tree (MSTP)             |
| IEEE 802.1p                    | QoS Priority Marking                      |
| IEEE 802.1Q                    | VLANs                                     |
| IEEE 802.1v                    | VLAN Classification                       |
| IEEE 802.1ad                   | Q-in-Q                                    |
| IEEE 802.1X                    | Port Security                             |
| IEEE 802.3AB                   | LLDP                                      |
| IEEE 802.3ah                   | OAM                                       |
| IEEE 802.1ag                   | Connectivity Fault Management             |
| IEEE 802.3at                   | 30W PoE+                                  |
| IEEE 802.3bt                   | 60 & 90W PoE                              |
| RFC112, 2236, 3376, 4604, 5711 | IGMP v1, v2, v3                           |
| RFC2236, 3376                  | IGMP Snooping                             |
| RFC8907                        | TACACS                                    |
| RFC2865, 2866                  | RADIUS                                    |
| RFC5424                        | Syslog                                    |
| RFC4250 - 4254                 | SSH                                       |
| RFC5246                        | TLS1.2 / HTTPS                            |
| RFC854                         | Telnet                                    |
| RFC2030                        | SNTP                                      |
| RFC2131                        | DHCP                                      |
| IEEE 1588-2008 v2              | Precision Time Protocol (PTP)             |
| IEC 62439-2                    | Media Redundancy Protocol                 |
| ITU-T G.8032                   | Ethernet Ring Protection Switching (ERPS) |
| ITU-T Y.1731                   | Performance Monitoring                    |
| Dante®                         | Audio Network Support                     |

## QoS.

|                  |                                |
|------------------|--------------------------------|
| Class of Service | 802.1p QoS & DSCP              |
| Diffserv         | RFC2474                        |
| Rate Limiting    | Ingress / Egress               |
| Priority Queue   | WRR / Strict / Hybrid Priority |

## Hardware Features.

|                   |                                                                                   |
|-------------------|-----------------------------------------------------------------------------------|
| Architecture      | Store-and-Forward                                                                 |
| Switch Latency    | <7µs                                                                              |
| Switch Fabric     | 54 Gbps (Non-Blocking Wire Speed On All Ports. 99.999% Error Free Data Integrity) |
| Address Table     | 8K MAC Entries                                                                    |
| Buffer Memory     | 4M bits                                                                           |
| Jumbo Frames      | 9.6K Bytes                                                                        |
| CPU               | 416MHz                                                                            |
| SDRAM             | 2Gb                                                                               |
| Flash             | 512Mb                                                                             |
| VLAN's            | 4K                                                                                |
| IGMP Groups       | 1024                                                                              |
| IPv6 MLD Groups   | 1024                                                                              |
| Throughput        | 40.18Mpps @ 64 bytes                                                              |
| Priority Queues   | 8                                                                                 |
| Bandwidth Control | Ingress Packet Filter and Egress Rate Limit                                       |

## Layer 3 Features.

|                 |               |
|-----------------|---------------|
| Static Routing: |               |
| Interfaces      | 8 Max†        |
| Routes          | 32 Max†       |
| DHCP            | Server (IPv4) |

†Software Routing. Consult AMG Technical Services For Application Support

## Software Features.

|                |                              |      |
|----------------|------------------------------|------|
| Redundancy     | STP                          | RSTP |
|                | MSTP                         | MRP  |
|                | ERPS (G.8032)                |      |
| VLAN           | 802.1Q                       |      |
|                | Port Based VLAN              |      |
|                | Private VLAN                 |      |
|                | Voice VLAN                   |      |
| MVR            | Multicast VLAN Registration  |      |
| MRP/GVRP       | VLAN Registration            |      |
| LACP           | Dynamic Trunk                |      |
|                | Static Trunk                 |      |
| SFP Monitoring | DDM                          |      |
| IGMP Snooping  | v1/v2/v3 (8 VLAN's Max)      |      |
| IGMP Querier   |                              |      |
| MLD Snooping   | IPv6 v1 (8 VLAN's Max)       |      |
| MLD Querier    | IPv6                         |      |
| IPMC           | 64 Profiles (128 Rules Each) |      |
| PTP            | Peer-to-Peer Transparent     |      |
|                | End-to-End Transparent       |      |

# Specifications.

## Security.

|                                           |                          |
|-------------------------------------------|--------------------------|
| Port Security                             | MAC/IP Based             |
| Storm Control                             | Rate Limiting            |
| 802.1x                                    | RADIUS Authentication    |
| Dynamic ARP Inspection                    |                          |
| Sticky MAC                                |                          |
| TACACS+                                   |                          |
| HTTPS/SSL                                 |                          |
| BPDU Guard                                |                          |
| DHCP Snooping                             |                          |
| Loop Protection                           |                          |
| IP Source Guard                           | IPv4 & IPv6              |
| IP Authorisation Managers                 |                          |
| Access (Policy) Control List (ACL L2/3/4) |                          |
| Custom User Rights                        | 15 Levels (20 Users Max) |

## Alarms/Contact Closures.

|               |                               |
|---------------|-------------------------------|
| Inputs        | 1x Dry Input                  |
| Outputs       | 1x Form A Relay               |
| Alarm Outputs | 1x Form A Relay               |
| Output Rating | 400V <sub>DC</sub> @ 0.1A Max |

## PoE Management.

|                                             |  |
|---------------------------------------------|--|
| Scheduling                                  |  |
| Ping Watchdog with Reboot                   |  |
| Enable/Disable, Priority Level, Power Level |  |

## Management.

|                              |                                                 |
|------------------------------|-------------------------------------------------|
| DHCP                         | Client / Relay (IPv4 & IPv6)<br>Option 66/67/82 |
| Event/Error Log              | Syslog Client                                   |
| Management Access            | SNMP<br>Web GUI<br>Telnet / SSH v2.0 / CLI      |
| Access Management            | Filtering                                       |
| SNMP                         | v1/v2c/v3                                       |
| RMON                         | 1/2/3/9 Groups                                  |
| Port Mirroring               |                                                 |
| Software Update              | HTTP/HTTPS                                      |
| Config Export /Import        |                                                 |
| Dual Firmware Images         |                                                 |
| FTP/TFTP/SCP/SFTP            |                                                 |
| SNTP                         | Client (IPv4 & IPv6)                            |
| Configuration                | IPv4 & IPv6                                     |
| LLDP                         | Link Layer Discovery Protocol                   |
| LLDP-MED                     |                                                 |
| sFlow                        |                                                 |
| Time Zone & Daylight Savings |                                                 |
| Cable Diagnostics            |                                                 |

## Interface.

|                                                                         |                                                                                                                       |
|-------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| LED Indicators                                                          | 2x Power<br>Fault<br>SFP Link/Activity<br>RJ45 Link/Activity<br>PoE (PoE Models Only)                                 |
| RJ45 Ports                                                              | 8/12/16x 10/100/1000T(X) RJ45<br>with Auto MDI/MDI-X and<br>1.5 kV Isolation Protection<br>Half & Full Duplex Support |
| SFP Slot                                                                | 2x 100M/1G/2.5G SFP and<br>2, 6 or 10x 100M/1G SFP                                                                    |
| Power, I/O, Alarm<br>Serial Console                                     | 1x 8-Way Screw Terminal<br>USB Type C                                                                                 |
| (All Ports Comply With EIA Ethernet Data<br>Communication Requirements) |                                                                                                                       |

## Power.

|                    |                                                     |
|--------------------|-----------------------------------------------------|
| Power Inputs       | 2                                                   |
| Operating Voltage: |                                                     |
| Non-PoE Models     | 12-56V <sub>DC</sub>                                |
| 30W PoE Models     | 50-56V <sub>DC</sub>                                |
| 90W PoE Models     | 52-56V <sub>DC</sub>                                |
| Power Consumption  | 15 Watts Max (without PoE Load)                     |
| Total PoE Budget   | 360W Max (model dependent)                          |
| PSE Modes          | Mode A (30W Ports)<br>Mode A, Mode B (60/90W Ports) |
| PoE Enabled Ports  | Ports 1-8, 1-12, 1-16 (model dependent)             |
| Protection         | Reverse Polarity<br>Overload                        |

## Mechanical.

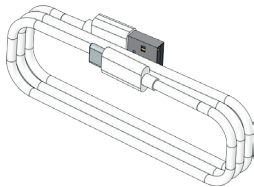
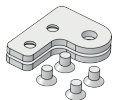
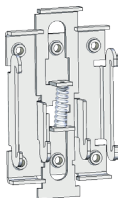
|                             |                                                                                             |
|-----------------------------|---------------------------------------------------------------------------------------------|
| Housing                     | Anodised Aluminium                                                                          |
| Dimensions:                 | (W x D x H)                                                                                 |
| Excluding DIN & Wall Mounts | 80 x 123 x 156 mm<br>3.15 x 4.84 x 6.14 in                                                  |
| IP Rating                   | IP40                                                                                        |
| Installation                | Wall Mount or DIN-Rail                                                                      |
| Construction                | All Parts & Conductive<br>Surfaces Are Non-Corrosive<br>Materials<br>No Self-Tapping Screws |
| Weight                      | 1.44kg / 3.17lb                                                                             |

# Specifications.

| Environmental.               |                                 | Regulatory.                            |                                   |
|------------------------------|---------------------------------|----------------------------------------|-----------------------------------|
| Operating Temp:              | -40 to +75°C / -40 to +167°F    | Safety                                 | IEC/EN 62368-1                    |
| 40 LFM Vented Enclosure      | -40 to +70°C / -40 to +158°F    | EMI                                    | EN55032 Class A                   |
| Sealed Enclosure             | -40 to +60°C / -40 to +140°F    |                                        | FCC Part 15B Class A              |
| 200 LFM Fan/Blower Enclosure | -34 to +75°C / -29 to +167°F    | EMS                                    | EN61000-4-2 (ESD)                 |
| Storage Temp.                | -40 to +85°C / -40 to +185°F    |                                        | EN61000-4-3 (RS)                  |
| Humidity                     | 5% to 95% (non-condensing)      |                                        | EN61000-4-4 (EFT)                 |
| MTBF                         | >500,000 hours                  |                                        | EN61000-4-5 (Surge)               |
| MTBF Standard                | Telcordia SR-332 GF 30°C        |                                        | EN61000-4-6 (CS)                  |
| Heat Dissipation             | 51 BTU/h (Non-PoE Models)       |                                        | EN61000-4-8 (PFMF)                |
|                              | 870 BTU/h (with 240W PoE Load)  | Shock                                  | IEC 60068-2-27                    |
|                              | 1280 BTU/h (with 360W PoE Load) | Free Fall                              | IEC 60068-2-32                    |
| Cooling                      | Passive Cooling                 | Vibration                              | IEC 60068-2-6                     |
| Noise Level                  | 0 dBA                           | Environmental                          | Reach                             |
|                              |                                 |                                        | RoHS                              |
|                              |                                 |                                        | WEEE                              |
| Packaging.                   |                                 | Supply Chain                           | NDAA & TAA Compliant              |
|                              |                                 |                                        | BABA Compliant (Order -B Version) |
| Shipping Weight              | 1.66kg / 3.66lb                 |                                        |                                   |
| Dimensions                   | (W x D x H)                     |                                        |                                   |
|                              | 260 x 200 x 84 mm               |                                        |                                   |
|                              | 10.24 x 7.87 x 3.31 in          | Designed to meet NEMA TS2 & EN 50121-4 |                                   |

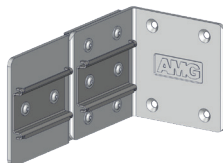
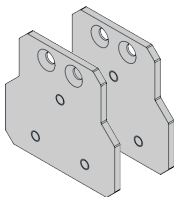
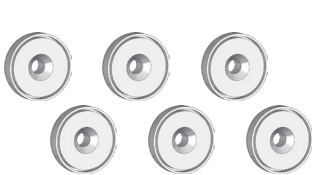
# Included Accessories.

|                        |                                                                                        |
|------------------------|----------------------------------------------------------------------------------------|
| DIN Rail Adapter       | Rear Mounted Metal DIN Rail Clip & Screws For DIN Rail Mounting AMG570 Series Products |
| Wall Mounting Brackets | 2x Wall Mounting Brackets & Screws For Wall / Surface Mounting AMG570 Series Products  |
| USB Console Cable      | USB Type A to Type C Console Cable (1.5M)                                              |



# Optional Accessories.

|               |                                                                                                                                                                                                      |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AMGMNT-MAG-06 | 6x Rear Mounted Magnets, Mounting Plates & Screws Kit For AMG570 Series, need to be ordered separately                                                                                               |
| AMG2035       | Side Mounted Wall Bracket Adapter Kit For DIN Rail Mounting AMG570 Series Products In Depth Restricted Installations.                                                                                |
| SFP Modules   | Adapter Kit Can Be Extended Using The Included Extension Kit To Also Mount A DIN Rail Power Supply Up To 480W<br>100Mb & 1Gb Optical/Copper Modules see separate list, need to be ordered separately |



# Part Numbers.

## Industrial Layer 2+ Managed 16 Port Switches (8+8)

|                           |                                                                                |
|---------------------------|--------------------------------------------------------------------------------|
| AMG570-8G-8S*             | 8x 10/100/1000TX, 6x 100M/1G SFP, 2x 100M/1G/2.5G SFP                          |
| AMG570-8GAT-8S-P240*      | 8x 10/100/1000TX (8x 30W PoE), 6x 100M/1G SFP, 2x 100M/1G/2.5G SFP             |
| AMG570-2GBT-6GAT-8S-P360* | 8x 10/100/1000TX (2x 60/90W & 6x 30W PoE), 6x 100M/1G SFP, 2x 100M/1G/2.5G SFP |

## Industrial Layer 2+ Managed 20 Port Switches (16+4 & 12+8)

|                              |                                                                                             |
|------------------------------|---------------------------------------------------------------------------------------------|
| AMG570-16G-4S*               | 16x 10/100/1000TX, 2x 100M/1G SFP, 2x 100M/1G/2.5G SFP                                      |
| AMG570-16GAT-4S-P360*        | 16x 10/100/1000TX (16x 30W PoE), 2x 100M/1G SFP, 2x 100M/1G/2.5G SFP                        |
| AMG570-4GBT-8GAT-4G-4S-P360* | 16x 10/100/1000TX (4x 60/90W, 8x 30W PoE & 4x Non-PoE), 2x 100M/1G SFP, 2x 100M/1G/2.5G SFP |
| AMG570-12G-8S*               | 12x 10/100/1000TX, 6x 100M/1G SFP, 2x 100M/1G/2.5G SFP                                      |
| AMG570-12GAT-8S-P360*        | 12x 10/100/1000TX (12x 30W PoE), 6x 100M/1G SFP, 2x 100M/1G/2.5G SFP                        |
| AMG570-4GBT-8GAT-8S-P360*    | 12x 10/100/1000TX (4x 60/90W & 8x 30W PoE), 6x 100M/1G SFP, 2x 100M/1G/2.5G SFP             |

## Industrial Layer 2+ Managed 24 Port Switches (16+8)

|                              |                                                                                             |
|------------------------------|---------------------------------------------------------------------------------------------|
| AMG570-16G-8S*               | 16x 10/100/1000TX, 6x 100M/1G SFP, 2x 100M/1G/2.5G SFP                                      |
| AMG570-16GAT-8S-P360*        | 16x 10/100/1000TX (16x 30W PoE), 6x 100M/1G SFP, 2x 100M/1G/2.5G SFP                        |
| AMG570-4GBT-8GAT-4G-8S-P360* | 16x 10/100/1000TX (4x 60/90W, 8x 30W PoE & 4x Non-PoE), 6x 100M/1G SFP, 2x 100M/1G/2.5G SFP |

## Industrial Layer 2+ Managed 24 Port Switches (12+12)

|                             |                                                                                  |
|-----------------------------|----------------------------------------------------------------------------------|
| AMG570-12G-12S*             | 12x 10/100/1000TX, 10x 100M/1G SFP, 2x 100M/1G/2.5G SFP                          |
| AMG570-12GAT-12S-P360*      | 12x 10/100/1000TX (12x 30W PoE), 10x 100M/1G SFP, 2x 100M/1G/2.5G SFP            |
| AMG570-2GBT-10GAT-12S-P360* | 12x 10/100/1000TX (2x 60/90W, 10x 30W PoE), 10x 100M/1G SFP, 2x 100M/1G/2.5G SFP |

\* For conformal coated circuit boards please add /C to the end of the part code. For example AMG570-8GAT-8S-P240/C



Standard Part Codes Are Manufactured In The UK. For Made In The USA Models That Are Fully BABA Compliant Add **-B** At The End Of The Partcode.  
e.g. AMG570-16GAT-4S-P360-B

# Recommended PSUs.

## Non-PoE Models

|                 |                                                                                                                   |
|-----------------|-------------------------------------------------------------------------------------------------------------------|
| AMGPSU-W12-P25  | Plug Top Mounting Lite Industrial Grade PSU, 0 to +70°C, 12VDC, 25W, NDAA Compliant, UK/EU/US Plug Heads Included |
| AMGPSU-I12-P24A | DIN-Rail Mount Industrial Grade PSU, -25 to +70°C, 12VDC, 24W, NDAA Compliant*                                    |
| AMGPSU-T12-P15  | DIN-Rail Mount Industrial Grade PSU, -25 to +70°C, 12VDC, 15W, NDAA & TAA Compliant*                              |

## PoE Models

|                  |                                                                                             |
|------------------|---------------------------------------------------------------------------------------------|
| AMGPSU-I48-P240A | DIN-Rail Mounting Industrial Grade PSU, -40 to +70°C, 48-55VDC, 240W, NDAA Compliant**^     |
| AMGPSU-I48-P290  | DIN-Rail Mounting Industrial Grade PSU, -25 to +70°C, 48-55VDC, 290W, NDAA & TAA Compliant* |
| AMGPSU-I48-P480A | DIN-Rail Mounting Industrial Grade PSU, -40 to +70°C, 48-55VDC, 480W, NDAA Compliant**^     |
| AMGPSU-I48-P530  | DIN-Rail Mounting Industrial Grade PSU, -25 to +70°C, 48-55VDC, 530W, NDAA & TAA Compliant* |

\* Also available in kit form including mains line cord, DC cable and DIN rail. Order with -K at the end of the part code (e.g. AMGPSU-I48-P240A-K).

^ Also available with no exposed mains terminals and direct IEC input kit. Order with -IEC at the end of the part code (e.g. AMGPSU-I48-P240A-IEC).

Dante® is a registered trademark of Audinate Pty Ltd. Audinate does not endorse or sponsor this product.

In a continuing effort to improve and advance technology, product specifications are subject to change without notice. Please visit [www.amgsystems.com](http://www.amgsystems.com) for the latest product specifications.

Proud to be a UK & USA  
Manufacturer